Bioinformatics-Guided Comparative Analysis of Moringa and Tamarind for Microplastic Removal

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Abstract

Microplastic pollution is one of the biggest problems affecting the environment and, consequently, the human population. This problem leads to several disorders in humans, such as endocrine disruption, inflammation, and organ damage. Microplastics' small size makes them resistant to conventional water treatment methods, with widespread contamination in water bodies around the world. This project investigates the use of two well-known natural coagulants, such as Moringa and tamarind seed extracts, to clump and remove microplastics from water and compare the efficiency of both. The computational approach was used, including molecular docking, interaction analysis, and molecular descriptor analysis, to evaluate the potential direct interactions of the *Moringa oleifera* coagulant protein (MOCP) receptor and tamarind polysaccharides with representative microplastic surrogates such as PP, PE, PS, and PET, using Autodock Vina and SMINA, respectively. Unlike approaches that examine protein-based coagulation or polysaccharide-based flocculation separately, this study compares MOCP and tamarind xyloglucan components within a single computational framework, allowing their distinct predicted microplastic-interaction mechanisms to be evaluated side by side before experimental validation. To validate the docking results, interatomic distances were measured in PyMOL. SwissADME was used to determine the physicochemical characteristics of the molecules. It was identified that MOCP performed as a better coagulant than tamarind for the MPs tested. MOCP had a higher binding affinity to PET and PS as compared to PP and PE. Molecular descriptor analysis for tamarind revealed high hydrophilicity and hydrogen bonding capacity with xyloglucan components, and the seed's lipid molecule showed chemically distinct properties. Thus, this study is a useful direction for using plant-based products which are readily available, biodegradable, and low-cost, making them a suitable alternative to artificial removal compounds for microplastics from the water.

Keywords

Microplastic Removal, Molecular Docking, Moringa, Tamarind, Bioinformatics.

Introduction

Plastic pollution is one of the biggest global issues of the 21st century. Humans generated only 2 million tons of plastic waste in 1950. This number has exponentially increased to over 450 million tons of plastic waste 75 years later in 2025.[1] This trend is primarily due to increasing populations, industrialization, and rising consumer needs, posing a major threat to both terrestrial and aquatic ecosystems due to a lack of appropriate disposal mechanisms. Plastics are extremely durable and never completely disappear from

the environment; instead, they break down into smaller plastic particles called microplastics. Microplastics are extremely small pieces of plastic, smaller than 5 millimeters. They result from the degradation of larger pieces of plastic and can often be found in water ecosystems. Every liter of water can contain thousands of microplastic particles. Even filtered and bottled water can often contain significant quantities of microplastics. Its concentration in natural water bodies ranges from 1×10^{-2} to 10^{8} pieces per m^3. [2] Microplastics in water affect both marine life and humans. After being ingested, microplastics are bioaccumulated in the fat cells of organisms and don't get flushed out. These can be extremely harmful for humans, causing a variety of diseases: types of cancer, respiratory disorders, inflammatory bowel disease, and cell damage. [3] The lack of safe drinking water causes severe problems to humans and hence calls for more extensive research into solutions for this.

Existing research on this subject includes various techniques to remove microplastics from drinking water. A common technique to remove MPs is coagulation.[4] This research paper reviews several coagulation-based removal methods and focuses specifically on microplastics in aquatic ecosystems. It provides a foundation for coagulation processes using natural coagulants, making it highly relevant and comprehensive. The results from this research paper indicate that coagulation is an effective process and can be useful in removing MPs, but it is dependent on MP size and the conditions created before coagulation. Other research papers build on these findings, focusing on optimizing this removal process. [5] It is aimed towards the same research area and provides further depth through quantitative results by evaluating the advantages and disadvantages of using moringa as a plant-based coagulant and conducting simulations using SEM imaging. The key finding that this research paper adds value through is that instead of a raw process, a hybrid coagulation using two or more coagulants (chemical and natural) at the same time. Getting even more specific, other papers build on the foundations already established by the previously discussed investigation. [6] It provides value through highly targeted research, prioritizing the influence of a few factors on the efficiency of MP removal. These factors include coagulant dose, conditions of the water, such as pH, and the MP particle size, which are significant factors in the efficiency of removal. This research discusses how the process can be enhanced by conducting jar tests, thus providing valuable insights. These research studies provide a baseline for optimization and for conducting a more focused study.

Other studies narrow down their focus specifically on plant-based coagulants and moringa oleifera seed extracts, and further evaluate moringa through more trials and calculations. [7] The key finding of this study is that moringa effectively reduces chemical load, acting as a green water treatment technology. It compares moringa to other chemical coagulants, alum, and PAC. The paper emphasizes sustainability, showing the benefits of employing natural coagulants. The next research paper doesn't include moringa. [8] However, it explains the physical removal process in detail and discusses flocculation instead of coagulation. It makes use of microscopy to show how MPs can be removed through polymer bridging, providing a mechanistic foundation. Furthermore, another paper unearths different perspectives and facets of the research.[9] Its key value addition comes from the finding that removal efficiency is directly proportional to polymer concentration. It uses methods such as optical counting, turbidity analysis, and quantitative analysis to compare polymers. This helps underscore dosage levels. On the other hand, other papers provide a broader perspective on moringa's general purification properties and the various water treatment applications of moringa. [10] It uses protein extraction and sedimentation analysis techniques to provide a protein-based explanation of the process. It explains how moringa proteins neutralize the effective charge of microplastics. Lastly, a particular investigation makes use of bioinformatics, simulations, and dry lab tests, contrasting with wet lab experimentation conducted by other studies. [11] It identifies the parts of moringa responsible for this coagulation process down to the specific proteins by focusing on seed protein-particle interactions. It evaluates moringa's effectiveness through molecular modeling technology.

Although these research papers contain thorough and credible investigations, they do have limitations, which come in 4 main types. Firstly, certain research papers limit their studies to specific coagulants. While some only discuss natural coagulants, others only explore synthetic coagulants. Moringa is widely regarded as an effective coagulant; however, some research papers don't include it in their research. Secondly, research is limited by experimental conditions. Some studies conduct experimentation only in idealistic laboratory conditions, neglecting real-world implications. Others lack a large sample size or only conduct

short-term experimentation. Thirdly, some research wasn't focused solely on MPs. Certain papers evaluated the generic water purification capability of coagulants, while others tested turbidity instead of the level of MPs. Lastly, the papers lack variety in their method of study. Most use wet lab experiments, with only one paper employing molecular modeling software to conduct dry lab experimentation. These limitations in previously conducted studies leave opportunities for further research to be conducted and to build on this topic. There is scope to include testing across different plastic types (PVC, nylon, PET, etc.). Along with this, testing can be conducted in natural water bodies rather than lab conditions, and focus on optimizing pH, dosage, and contact conditions as well. Avenues for further research include the investigation of health problems arising from moringa residue and the use of modeling software to conduct experimentation due to the lack of this approach in existing work. Thus, this project aimed to investigate the use of natural coagulants like Moringa seed extracts and plant starches to clump and remove microplastics from water. The removal process will be tested using microscopy or open-source image analysis tools. The project aims to use readily available, biodegradable materials to devise microplastic removal methods that can be scaled. This addresses an important but understudied environmental problem. This method has several applications, including rural homes, schools, or small-scale community filtration projects.

Methods

Data collection and processing

The first step of undertaking the research was data collection. This involved research to find various types of datasets, statistical figures, and qualitative information. The 2 primary focus areas were microplastics and biopolymers. For microplastic research, this study explored regional MP levels, global MP distribution maps, tables with statistical data, images showing the structure of MPs, and MP composition and chemical characteristics. Additionally, the different types of MPs, such as PP, PE, PET, etc., were identified and differentiated. The analysis revealed that PP and PE dominate the sample, both non-polar, low-density polymers that are hydrophobic and resist conventional water treatment.

This matters because Moringa's cationic proteins work via charge neutralization, which is more effective on charged or polar particles. PP and PE fragments may require surface pre-treatment or higher coagulant doses. The presence of diverse polymer types also means removal efficiency will vary across the sample. For the biopolymers, this study researched the chemical and biological composition of 2 different plant-based sources: Moringa and Tamarind. Specifically, the protein/polysaccharide type and molecular structure were studied in detail to identify how the plant polymer helps in the filtration of MPs. This data was collected and then processed by filtering out information most relevant in the context of this study. Moringa's active component is a low molecular weight cationic protein (6-16 kDa) that works via charge neutralization. Tamarind's active component is a high molecular weight neutral polysaccharide (500-3000 kDa) that works via bridging flocculation and hydrogen bonding. Moringa targets charged/polar surfaces, Tamarind targets hydrophobic surfaces through physical entrapment. Together, they cover different removal mechanisms.

Molecular Docking

To analyze the compatibility of both plant-based compounds with the different types of microplastics, this study made use of a bioinformatics technique known as molecular docking. This process primarily made use of a website called Tamarind Bio to automatically conduct the analysis. The process started with finding files with the structure of the MOCP protein in Moringa and the molecular structure of the MP. The structures of the microplastics were drawn using ChemDraw 8.0, an industry-standard software used to make scientific drawings of molecular structures. Along with this, the PyMol 3.1.8 software application was used to convert files into the right format suitable for uploading into Tamarind Bio, which accepted SDF and PDB files. After the files were uploaded, the physical molecular structures were calibrated and prepared for analysis for both 8 and 15 exhaustiveness to find the optimal binding position. The analysis was then automatically conducted by the Autodock Vina tool in Tamarind Bio, within a few minutes. This gave a result in the form of binding affinity as shown in Table 1. For each docking simulation, the top-ranked pose with the lowest

predicted binding energy returned by the docking engine was selected. The binding affinity of this pose was reported in the results tables and, for the MOCP complexes, the same selected pose was used for subsequent PyMOL interaction-distance analysis. A similar process was followed for the docking of tamarind. MOCP receptor PDB files were uploaded to Tamarind Bio on the SMINA software. After calibrating the docking box to the structure, SDF MP ligand files were also uploaded. This process was followed for a few components of Tamarind: glucose, xylose, galactose, and cellobiose. After running the software with combinations of these molecules with each of the MPs, their binding affinity was collected and recorded in Table 4.

Interaction Analysis

Along with molecular docking, this study used interaction analysis to determine how efficient the moringa proteins are in coagulating ligands. To carry out this process, this study used PyMOL. First, the docked PDB files from Tamarind Bio were exported to PyMOL to visualize the ligand and protein. The software allows the docked structure to be displayed in various formats, such as sticks, lines, surface, etc., which provide useful visualizations of the protein as seen in figures 1 and 2. Next, the measurement wizard tool was used to find the distances between the microplastic ligand and the protein after displaying the structure in stick format Using the PyMOL Measurement Wizard, the five shortest interatomic distances between atoms of the docked microplastic surrogate and MOCP were measured for each complex and recorded in Å. The same five-distance criterion was applied to the PS, PP, PE, and PET complexes to maintain consistency across comparisons.

Molecular Descriptor Analysis

The data collection for tamarind polysaccharides was conducted by comparing tamarind seeds with several other molecules based on various parameters. The molecules tamarind was compared to were glucose, xylose, galactose, and cellobiose. The process started by accessing PubChem to find the SMILES (Simplified Molecular-Input Line-Entry System) codes for all these molecules. Then, SwissADME was run with those SMILE codes, and the molecules' relevant characteristics were extracted. The molecules were analyzed across physicochemical and pharmacokinetic parameters. Molecular formula and weight describe basic composition and size. Heavier atoms are characteristic of more complex structures. H-bond donors and acceptors indicate the ability for hydrogen bonding to take place, which is important for surface interactions. Topological Polar Surface Area (TPSA) measures the polar surface area, indicating hydrophilicity. Rotatable bonds describe molecular flexibility. The logarithm of the Partition coefficient (LogP) measures hydrophobicity, with negative values indicating water affinity and positive values indicating lipophilicity. Log S and water solubility class describe aqueous solubility. Bioavailability score estimates interaction readiness in aqueous systems. Synthetic accessibility scores how easily the molecule can be isolated or produced. These data can be seen recorded in Table 3.

Results

Molecular Docking Analysis for Moringa

The molecular docking yielded results for 8 combinations. Each of the 4 most common microplastic types bonds with the MOCP protein with an exhaustiveness of 8 or 15. The compatibility of the protein was determined by binding affinity, showing how well each plant-based coagulant worked with different types of MPs.

Table 1. The binding affinity of MOCP with different types of MPs at 8 and 15 exhaustiveness.

Biopolymer Protein	**Microplastic Ligand**	**Binding affinity (Kcal/mol) Exhaustiveness - 8**	**Binding affinity (Kcal/mol) Exhaustiveness - 15**
Moringa MOCP	Polystyrene (PS)	-4.621	-4.618

Moringa MOCP	Polypropylene (PP)	-2.154	-2.175
Moringa MOCP	Polyethylene (PE)	-2.598	-2.611
Moringa MOCP	Polyethylene Terephthalate (PET)	-4.958	-5.015

The data in Table 1 shows that the moringa MOCP protein is most compatible and works as the best coagulant with PET MPs when run at 15 exhaustiveness and with PS MPs when run at 8 exhaustiveness. These have the highest binding affinity at -5.015 Kcal/mol for PET and -4.621 Kcal/mol for PS. Binding affinity is an indicator for MP aggregation because it measures the strength of non-covalent interactions between the MOCP protein and the microplastic surface. A more negative value indicates a thermodynamically more stable complex, meaning the protein binds more readily and remains attached longer. This promotes surface coating and charge neutralization, both prerequisites for particle aggregation and flocculation. Molecular docking is an effective bioinformatics technique to use here for several reasons. Firstly, the molecular structures of the MOCP protein and the different types of MPs are readily available in easily accessible databases. Secondly, using a software-based simulation approach before conducting primary data collection through hands-on experimentation is much easier and provides validation for the foundation of the research. Wet lab work, on the other hand, would require the study to isolate very specific chemicals and carry out jar tests, which are highly subject to uncertainties and environmental conditions. Lastly, it also provides objective, easily comparable results that can quantitatively identify compatibility for this study.

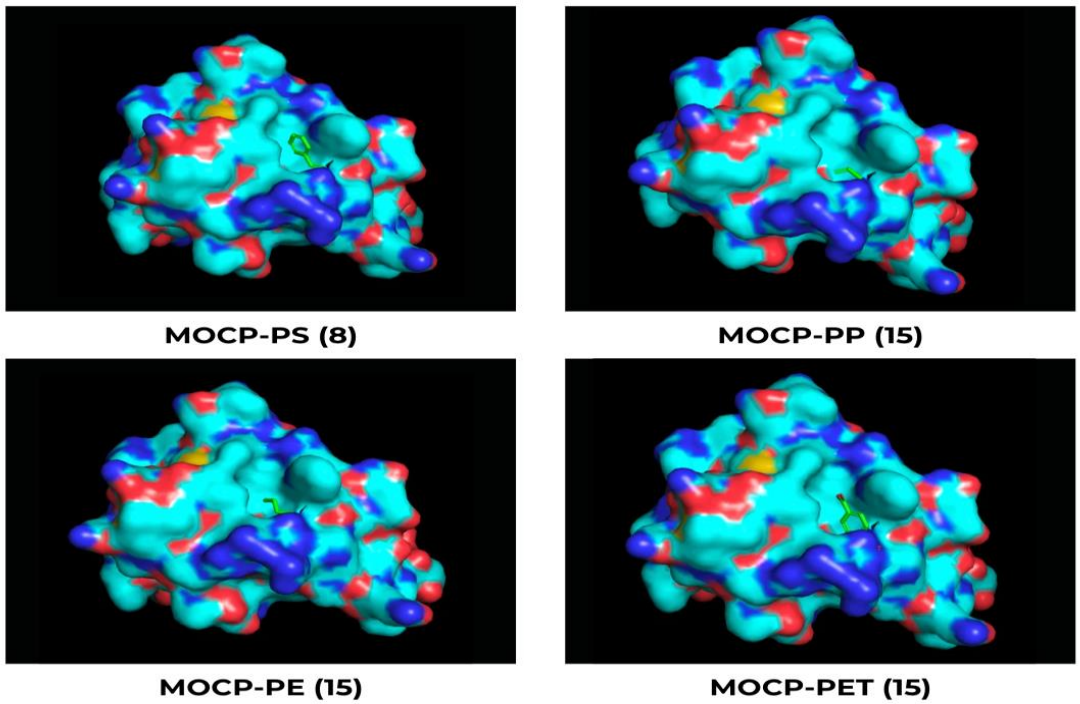


Figure 1. Surface Representation of the MOCP of Moringa with different microplastic ligands. It shows binding interactions of microplastics such as PS, PP, PE, and PET with surface representations of MOCP. The ligand is shown as sticks in green within the binding pocket. Color mapping of the surface indicates properties with distinct regions reflecting differences in polarity and binding affinity.

Interaction Analysis Results

Interaction analysis works together with molecular docking by providing a spatial, visual confirmation of binding success. On one hand, docking gives a single binding affinity score, while on the other, interaction analysis measures the actual distances between the closest atoms of the protein and ligand after docking. Shorter distances point towards stronger, more stable interactions. The PET combination in Table 2 shows the shortest distances by a big margin, aligned with its lowest binding affinity of -5.015 Kcal/mol at exhaustiveness 15. PS is next with small distances, again consistent with the second strongest binding affinity of -4.621 Kcal/mol. PP and PE show larger distances and weaker binding affinities, confirming the same trend. This cross-validation between two distinct analytical methods strengthens the reliability of the results and rigor of the experimental methodology. Finally, from this we can conclude that the MOCP protein has the strongest interaction with PET, followed by PS, PP, and PE.

Table 2. Bond distances of MOCP with different types of MPs.

Biopolymer Protein	**Microplastic Receptor**	**Distances**				
Moringa MOCP	Polystyrene (PS)	1.9	3.6	3.6	3.6	3.7
Moringa MOCP	Polypropylene (PP)	1.9	3.3	3.8	4	4.1
Moringa MOCP	Polyethylene (PE)	1.8	2.6	3.7	3.7	4.1
Moringa MOCP	Polyethylene Terephthalate (PET)	0.8	0.9	1.2	2	3.1

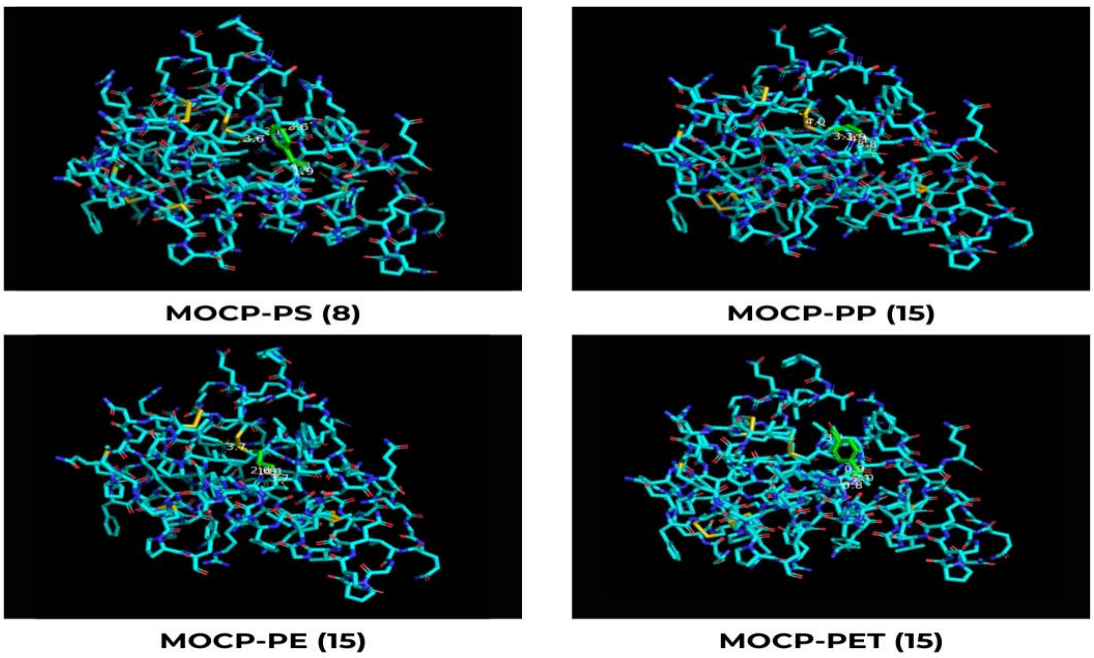


Figure 2. Stick representation of MOCP with MP receptors and binding distances. This shows a visualization of the position and orientation of MOCP, comparing each receptor environment, with labeled interaction distances between the closest points. Variations in proximity and interaction geometry show the differences in binding strength and molecular compatibility across the polymer types.

Molecular Descriptor Analysis

The data in Table 3 shows that all monosaccharides (glucose, xylose, galactose) were identified to be highly soluble and had negative LogP values ranging from -2 to -2.23. This confirmed the strong hydrophilicity of

the monosaccharides due to multiple hydroxyl groups. Tamarind xyloglucan's flocculation mechanism is underpinned by extensive hydrogen bonding capacity, which is indicated by its TPSA values (90-110 Å²) and high H-bond donor/acceptor values. Cellobiose follows a congruent pattern but at a larger scale: 23 heavy atoms, 8 H-bond donors with a TPSA of 189.53 Å² and a more negative LogS (-3.54 compared to approximately 1.1 for monosaccharides), showing lower, but still significant, solubility as chain length increases. The tamarind seed reading value is an outlier: zero H-bond donors/acceptors, a TPSA of 0, and a positive LogP of 3.62. It behaves like a hydrophobic, lipid-like molecule. It is representative of a fatty acid or waxy fraction of the seed coating instead of the active xyloglucan. This contrasts with other readings in Table 4 since all others are polar, water-soluble, and hydrogen-bonding. This differentiation is relevant as the hydrophobic seed fraction and the hydrophilic xyloglucan polysaccharide indicate two chemically inverse components within the same biopolymer.

Table 3. Molecular descriptor analysis for Components of the tamarind seed.

Molecule Name	H-bond Donors	H-bond Acceptors	TPSA (Å²)	Consensus LogP	Water Solubility Class
Glucose	5	6	110.38	-2.23	Highly Soluble
Xylose	4	5	90.15	-2	Highly Soluble
Galactose	5	6	110.38	-2.16	Highly Soluble
Cellobiose	8	11	189.53	-3.54	Highly Soluble
Tamarind Seed	0	0	0	3.62	Soluble

Docking Analysis for Tamarind

The molecular docking gave results for 16 combinations visible in Table 4. Each of the 4 most common microplastic types was docked against 4 tamarind monosaccharide units (glucose, xylose, galactose, cellobiose) representing the building blocks of tamarind xyloglucan.

Table 4. Bond Affinities of different components of Tamarind with different types of MPs.

Microplastic Receptor	Biopolymer Protein	Binding Affinity (Kcal/mol)
Polyethylene (PE)	tamarind_glucose	-0.6
Polyethylene (PE)	tamarind_xylose	-0.5
Polyethylene (PE)	tamarind_galactose	-0.6
Polyethylene (PE)	tamarind_cellobiose	-0.5
Polypropylene (PP)	tamarind_glucose	-0.5
Polypropylene (PP)	tamarind_xylose	-0.5
Polypropylene (PP)	tamarind_galactose	-0.5
Polypropylene (PP)	tamarind_cellobiose	-0.5
Polystyrene (PS)	tamarind_glucose	-1
Polystyrene (PS)	tamarind_xylose	-0.9
Polystyrene (PS)	tamarind_galactose	-1
Polystyrene (PS)	tamarind_cellobiose	-1.2

Polyethylene Terephthalate (PET)	tamarind_glucose	-1.5
Polyethylene Terephthalate (PET)	tamarind_xylose	-1.4
Polyethylene Terephthalate (PET)	tamarind_galactose	-1.5
Polyethylene Terephthalate (PET)	tamarind_cellobiose	-2.3

According to the data in Table 4, PET consistently has higher binding affinity values across all tamarind components, with the cellobiose and PET combination yielding the highest result of -2.3 Kcal/mol. Glucose and galactose follow when bonded to PET at a binding affinity of -1.5 with xylose at -1.4. From this, we can conclude the trend that larger, more structurally complicated disaccharide molecules have more stable interactions. The next strongest receptor can be identified as PS; once again, cellobiose performs the best with a -1.2 Kcal/mol binding affinity. PP and PE are both relatively weaker with binding affinities close to -0.5 Kcal/mol. This pattern confirms the theory that tamarind xyloglucan is better suited for aromatic and polar plastic surfaces, such as PET and PS, across all tamarind units as compared to non-polar polyolefins, like PP and PE.

Moringa vs Tamarind Performance

In terms of their overall performance, Moringa MOCP showed better results of binding affinity across all the microplastic types when compared with tamarind xyloglucan. Out of the 4 microplastics, PET yielded the best results with MOCP reaching a maximum 5.015 Kcal/mol and tamarind cellobiose reaching -2.3 Kcal/mol. Performance declined with PS and was significantly lower for PE and PP. Importantly, tamarind consistently showed close to -0.5 Kcal/mol binding for all units with PP, pointing towards uniform binding and minimal interaction. However, this still does not allow the research to conclude that Moringa is better in an absolute sense since both the biopolymers work using distinct mechanisms.

Moringa works via charge neutralization, making it more effective on polar and aromatic surfaces like PET and PS. At the same time, tamarind acts through hydrogen bonding and bridging flocculation, contributing physical entrapment that complements Moringa's charge-based action. The combined use of both biopolymers, therefore, offers broader-spectrum coverage than either coagulant alone.

Discussion

This research employed a variety of bioinformatics research methods to analyze Moringa Oleifera and Tamarind. The computational approach included the use of molecular docking, interaction analysis, and molecular descriptor analysis to evaluate the potential direct interactions with representative microplastic surrogates. The MOCP receptor was docked against 4 types of microplastics, PP, PE, PS, and PET, using the Autodock Vina software on tamarind Bio, while tamarind composite molecules, including glucose, xylose, galactose, and cellobiose, were docked using the SMINA software. To validate the docking results, interatomic distances were measured in PyMol. SwissADME was used to determine the physicochemical characteristics of the molecules present in tamarind. This study used a bioinformatics, dry lab approach to conduct an accurate comparison between the 2 biopolymers and provide the basis for experimental validation.

Several studies can be cross-referenced to confirm the accuracy of the approach and the effectiveness of Moringa as a coagulant. A study conducted by the University of Minnesota found that MOCP’s coagulating capabilities are like Alum, the most commonly used artificial substance. The study used jar tests to specifically analyze their performance when coagulating PE Microplastics. [12] Another study showed that MOCP had a removal rate of up to 92.6% for polyamide MPs. In fact, it showed that Moringa acted as a better coagulant than alum due to the charge neutralization and bridging mechanisms it makes use of. [13] The Oak Powder study used a similar Interaction analysis approach in its research to find the surface adsorption of tannin compounds, validating the effectiveness of this study's research methodology. [14]

Additionally, a literature review paper confirms this study's approach and bioinformatics techniques by showing how charge neutralization and bridging are the primary coagulation mechanisms that biopolymers use. [15]

The study uses a dry lab approach but does not confirm the findings through wet lab experimentation, meaning results are only indicative and may not be completely accurate. The use of a web interface software to perform docking limits the exhaustiveness and thoroughness with which research was conducted, leaving scope for more in-depth analysis using heavy computational servers. Tamarind was also docked as individual isomers instead of as a full xyloglucan chain, which may underestimate the bridging effects. A further limitation is the use of small molecular surrogates to represent complex polymers. The PE, PP, PS, and PET models capture only selected repeat-unit chemistry and do not reproduce polymer chain length, crystallinity, weathering, surface oxidation, or heterogeneous environmental coatings. Likewise, glucose, xylose, galactose, and cellobiose represent building blocks of tamarind xyloglucan rather than the complete high-molecular-weight polysaccharide. Consequently, the docking energies reported here should be interpreted as comparative molecular-interaction indicators rather than direct predictions of whole-particle microplastic removal efficiency. Future scope could include a more comprehensive analysis of tamarind to more precisely understand the coagulation performance. Additionally, the research could explore a variety of other microplastics and conduct wet lab tests to see results in real-world, instead of simulated, conditions.

Conclusion

To conclude, this study found that Moringa acts as a better coagulant than Tamarind, in most cases. It specifically showed very high binding to PET and significantly lower to PP and PE microplastics. Moringa contains proteins, while tamarind contains polysaccharides, which carry out its removal functions. However, both moringa and tamarind use complementary mechanisms: charge neutralization from Moringa and hydrogen bond bridging from tamarind can work together for wider microplastic removal. This study is useful as it uses polymer extracts from plants, which are readily available, biodegradable, and low-cost, making them a suitable alternative to artificial removal compounds like alum. They can be used for water treatment and microplastic removal in decentralized rural water systems where chemicals may not be accessible. From an application perspective, Moringa and tamarind are plant-derived and potentially locally obtainable materials, making them attractive candidates for decentralized coagulation-flocculation systems. However, practical deployment will require wet-lab optimization of extraction method, dose, pH, mixing and settling conditions, assessment of residual organic matter and treated-water quality, and subsequent pilot-scale validation before use in household or community water-treatment systems.

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